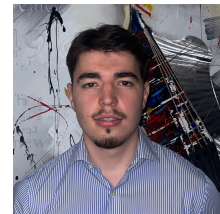


Joris Moczygeba

✉ joris.moczygeba@gmail.com ☎ +33 7 65 22 42 54
🌐 linkedin.com/in/joris-moczygeba 🐙 github.com/JorisM28



Computer Science Engineering Student - Specialization in AI & Cybersecurity at the National Graduate School of Engineering in Caen (ENSICAEN)

Final-year dual degree student in Computer Engineering and Management, specializing in Artificial Intelligence and Cybersecurity. Passionate about machine learning and data science, I am seeking a 6-month final-year internship (March – August 2027) to apply my analytical skills and develop scalable AI solutions.

Projects

- 2026 📌 **Sudoku Game** C++/Qt Desktop Application
- Designed the graphical user interface (UI) using Qt Designer, including an interactive grid and selection menu.
 - Implemented the game logic in C++ featuring multiple difficulty levels (Easy, Medium, Hard, Impossible) and a new game system.
- C++, QT, UI/UX, XML
- 2025 📌 **F1 Grand Prix - Optimal Trajectory** Algorithms
- Developed a C program to calculate the optimal racing trajectory under fuel constraints.
 - Implemented a mathematical solution to balance speed and fuel efficiency.
- C, ALGORITHMS, OPTIMIZATION
- 2024 📌 **Image Processing** Segmentation and Algorithms
- Implemented C algorithms to extract useful information by clustering pixels into homogeneous regions.
 - Prepared a technical report and presented the obtained results.
- C, IMAGE PROCESSING, DATA ANALYSIS

Professional Experience

- Since 09/26 📌 **AI Industrial Project - Lymphoma Diagnosis** Caen University Hospital (CHU), *Caen, France*
- Development and training of an AI algorithm based on a U-Net++ architecture with a ResNet50 encoder for pixel-by-pixel segmentation (benign, malignant, background) on digitized cytological slides.
 - Implementation of tile-based classification and ensemble prediction aimed at achieving diagnostic performance equivalent to or exceeding human expertise.
 - Dataset enrichment through image augmentation (rotation, cropping, brightness adjustments) and adaptation of the ingestion pipeline to the DICOM medical format.
 - Development of a graphical user interface (web application) to facilitate access and usability for doctors and technicians.
- PYTHON, PYTORCH, U-NET++, RESNET50, COMPUTER VISION, DICOM
- 04-08/2026 📌 **AI Research Internship** Czech Technical University, Artificial Intelligence Center, *Prague, Czech Republic*
- Implementation of a modular Machine Learning pipeline (K-P-V) applied to **chemical retrosynthesis**, on High-Performance Computing (HPC) clusters.
 - Development of an intelligent validation system using NLP models (ReactionT5) and chemoinformatics tools to filter AI hallucinations.
 - Execution of large-scale comparative benchmarks of advanced search algorithms (MCTS, Gaussian Processes).
 - Collaboration within an international research team using GitHub and agile methodologies.
- MACHINE LEARNING, HPC, NLP, REACTIONT5, MCTS, GITHUB

Education

- 2025 – 2027 📌 **Master in Management** – Dual Degree
EM Normandie Business School, Caen, France
- 2024 – 2027 📌 **Engineering Degree in Computer Science** – Major in AI & Cybersecurity
National Graduate School of Engineering in Caen (ENSICAEN), Caen, France
- Coursework: Artificial Intelligence, Cybersecurity, Data Science, Algorithms.
- 2021 – 2024 📌 **Preparatory Classes for French Engineering Schools (CPGE)** – PTSI/PT Track
Lycée Étienne Mimard, Saint-Étienne, France

Skills

Languages: Python, C/C++, Java, JS/TS, Dart

AI & Data: PyTorch, TensorFlow, Scikit-Learn, Generative AI

Systems & Tools: Linux/Unix, Bash, Git, SLURM (HPC)

Methodologies: Agile, Research Work, Analysis

Soft Skills: Critical thinking, Problem-solving, Team spirit

Web & Databases: React, Node.js, Flask, PostgreSQL, MySQL, MongoDB

AI Security: Adversarial ML, Model Robustness

Cybersecurity: IS Security, Cryptography

Languages: English (TOEIC 815), German (A2/B1), French (Native)

Other: Driver's License (Category B)

Interests

- Sports 📌 **Running** (regular practice) and **Physical conditioning**.
- Music 📌 **Piano** (8 years of practice).
- Travel 📌 Exploring Eastern Europe (Prague, Budapest, Warsaw).